



~~$$y = f(x)$$~~

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$$y_n = f(x_n)$$

$$m = f'(x_n)$$

$$\textcircled{1} \quad \frac{x - x_n}{x_{n+1} - x_n} = \frac{y - y_n}{y_{n+1} - y_n}$$

Eqn of line (tangent at x_n)

$$(x - x_n)(f'(x_n)) = y - f(x_n)$$

$$\textcircled{2} \quad y = 0 \text{ at } x = x_{n+1}$$

(Obtaining x_{n+1} in terms of $x_n, f'(x_n)$ and $f(x_n)$)

$$(x_{n+1} - x_n) f'(x_n) = 0 - f(x_n)$$

$\textcircled{3}$

$$x_{n+1} = x_n - \frac{f(x)}{f'(x_n)}$$